

Semester	Course Code	Course Category	Hours/ Week	Credits	Marks for Evaluation		
					CIA	ESE	Total
I	23UCS1CC1	CORE – I	5	5	25	75	100
Course Title		PROGRAMMING IN C AND C++					

SYLLABUS		
Unit	Contents	Hours
I	Overview of C – Importance of C – Basic Structure of C Programs – Constants – Variables – Data Types in C – Operators in C – Expressions – Managing Input and Output Operations – Decision Making and Branching – Various Forms of IF Statements – The Switch Statement – The ?: Operator – *The GO TO Statement*.	15
II	Decision Making and Looping – The WHILE Statement– The DO Statement – The FOR Statement – Arrays – Types of Arrays – Need for User-Defined Functions – The Form of C Functions – Category of Functions – Call by Value – *Call by Reference*.	15
III	Basic concepts of OOP – Structure of C++ Program – Operators and Data Types in C++ – Manipulators – Inline Functions – Default Arguments – *Recursion* – Function Overloading – Classes and Objects – Arrays of Objects – Objects as Function Arguments – Friendly Functions – Returning Objects.	15
IV	Constructors and Destructors – Constructors - Parameterized Constructors – Multiple Constructors in a Class – Copy Constructors – Destructors – Operator Overloading – Defining Operator Overloading – Overloading Unary Operators – Overloading Binary Operators – Overloading Binary Operators using Friends – Rules for Overloading Operators–Inheritance: Extending Classes – Defining Derived Classes – Single Inheritance – *Multilevel Inheritance* – Multiple Inheritance.	15
V	Pointers, Virtual Functions and Polymorphism – Pointers –Pointers to Objects –this Pointer – *Pointers to Derived Classes* – Virtual Functions – Pure Virtual Functions – Managing Console I/O Operations – C++ Streams – C++ Stream Classes – Unformatted I/O Operations – Formatted Console I/O Operations – Working with Files.	15
VI	Current Trends (For CIA only): Developing C/C++ coding for simple real world application problems	

..... Self Study

Text Book(s):
1. E. Balagurusamy, <i>Programming in ANSI C</i> , Tata McGraw Hill Education Private Ltd., New Delhi, Fifth Edition, 2011. 2. E. Balagurusamy, <i>Object Oriented Programming with C++</i> , Tata McGraw Hill Education Private Ltd., New Delhi, Fifth Edition, 2011.
Reference Book(s):
1. Yashavant Kanetkar, <i>Let Us C</i> , BPB Publications, New Delhi, Thirteenth Edition, 2013. 2. Bjarne Stroustrup, <i>The C++ Programming Language</i> , Addison-Wesley, New York, Third Edition, Eighth Impression, 2012.
Web Resource(s):
1. https://www.programiz.com/c-programming 2. https://www.geeksforgeeks.org/object-oriented-programming-in-cpp/ 3. https://onlinecourses.nptel.ac.in/noc22_cs40/preview 4. https://archive.nptel.ac.in/courses/106/105/106105151/ 5. https://nptel.ac.in/courses/106105151

Course Outcomes		
Upon successful completion of this course, the student will be able to:		
CO No.	CO Statement	Cognitive Level (K-Level)
CO1	Recall the basic concept of procedure and object-oriented programming	K1
CO2	Illustrate the fundamental definitions and concepts of C and C++ Programming	K2
CO3	Apply the concept of decision-making, looping, arrays, functions and OOP concepts	K3
CO4	Analyze various programming constructs of C and C++	K4
CO5	Evaluate and explain the suitable logic and principles of C and C++ Programming for solving real-time application problems	K5

Relationship Matrix:

Course Outcomes (COs)	Programme Outcomes (POs)					Programme Specific Outcomes (PSOs)					Mean Score of COs
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4	PSO5	
CO1	3	3	2	3	2	3	2	2	1	2	2.3
CO2	3	3	3	3	2	3	2	2	2	2	2.5
CO3	3	3	2	3	2	3	2	2	2	2	2.4
CO4	3	3	3	2	2	3	2	3	3	2	2.6
CO5	3	3	3	3	2	3	2	3	3	3	2.8
Mean Overall Score											2.52
Correlation											High

Mean Overall Score = Sum of Mean Score of COs / Total Number of COs

Mean Overall Score	Correlation
< 1.5	Low
≥ 1.5 and < 2.5	Medium
≥ 2.5	High

Course Coordinator: Dr. O.A. Mohamed Jafar